Class activity: inference for the regression slope

Group members:

Instructions: Work with a neighbor on the following activity. I will collect the handout at the end of class, and it will be part of your class participation grade. You will be graded only on effort – it is ok if you don't finish all the questions, or get them all correct.

Sparrow data

In this activity, you will practice hypothesis testing with a linear regression model. We will work with data on 116 sparrows, which were observed on Kent Island, New Brunswick. For each sparrow, we have information including:

- *Weight*: the weight of the sparrow (in grams)
- *WingLength*: the sparrow's wing length (in mm)

We fit the model

$$\text{wing length} = \beta_0 + \beta_1 \text{weight} + \varepsilon$$

Here is output from the fitted model:

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Coefficients:
            Estimate Std. Error
(Intercept)  8.75465    1.39601
Weight       1.31341    0.09756
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Questions

Researchers want to know whether there is a relationship between sparrow weight and wing length.

1. Write down null and alternative hypotheses corresponding to the research question.

2. Using the regression output, calculate the test statistic for the hypotheses in question 1.

3. Without calculating a p-value, do you think the observed test statistic would be unusual if the null hypothesis were true?

4. What will be the degrees of freedom for the t distribution used to calculate a p-value in this test?

5. Below is a picture of the appropriate t distribution. Indicate the area under the curve corresponding to the p-value for your hypothesis test.